TEXORA · LECTURE SHEET

Garments Washing

Wet and dry processes that give finished garments their hand feel, colour and fashion effects: normal, enzyme, stone and bleach washes, the dry-process family, and a fully dosed batch calculation.

AM-4101 · ADVANCED APPAREL MANUFACTURING-I · TOPIC 1 · SPECIALIZATION — APPAREL MANUFACTURING

1. Why garments are washed

Washing finished garments does four jobs: it **removes** sewing-stage contamination, starch and loose dye; it **softens** the fabric so the garment feels broken-in at first wear; it **stabilises** dimensions, taking out residual shrinkage before the customer does; and it **creates fashion effects** — fading, abrasion patterns and colour change that cannot be produced at fabric stage because they must follow the garment's seams and creases. Denim is the dominant washed product, but the same plant washes chinos, twill trousers and dyed garments. Processes divide into **wet processes** (in water, in washing machines) and **dry processes** (on the dry garment, before or between wet stages).

2. The principal wet processes

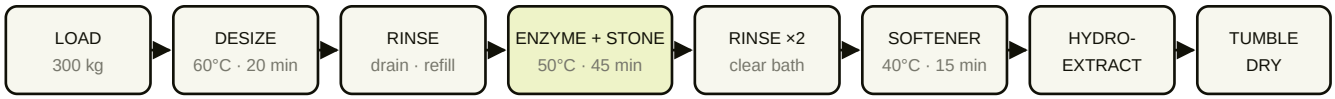
WASH	AGENTS AND ACTION	EFFECT OBTAINED
Normal (rinse) wash	Detergent and softener, mild temperature	Cleaning, softness, dimensional stability; no visible fading
Desizing	Enzyme (amylase) or oxidative agent removes weaving size/starch	Prepares denim for all later washes; without it, effects are uneven
Enzyme wash	Cellulase enzyme attacks surface cellulose fibres	Controlled fading and soft, clean surface; the standard denim wash
Stone wash	Pumice stones abrade the garment mechanically	Strong, irregular fading and worn look; heavier fabric and machine wear
Enzyme + stone	Both actions together at reduced stone load	Balanced fading with less damage — the industry workhorse

Bleach wash	Sodium hypochlorite or potassium permanganate oxidises indigo	Light to near-white shades; needs careful neutralising to protect fibre strength
Acid wash	Pumice soaked in bleach, tumbled with dry garments	Sharp white-on-blue marbled contrast
Pigment / tint wash	Pigment or tint applied in the wash bath	Over-dyed, vintage or colour-cast effects

3. The dry processes

Dry processes are worked on the dry garment, usually before the wet wash that fixes and blends them: **whiskers and creases** (patterns scraped or pressed at the hip and behind the knee), **hand sanding** (emery paper rubbed over thighs and seat), **destroying/grinding** (controlled damage at hems and pockets), **PP spray** (potassium permanganate sprayed for local bleach effect, then neutralised), **resin application** for permanent creases, and **tagging/clipping**. **Sandblasting** — once common for strong fading — is banned in many markets and buyer codes because silica dust causes silicosis in operators; a modern plant does not offer it, and this sheet treats it as history, not method.

4. A wash batch, calculated



Process sequence for the worked enzyme + stone wash batch

Figure 1 — Wet-process sequence. Times and temperatures are the recipe of §4; every wet wash is a variation on this skeleton.

WORKED EXAMPLE — DOSING A 300 KG ENZYME + STONE BATCH

Batch: **300 kg** of denim garments; liquor ratio **1 : 10** → bath = 300 × 10 = **3,000 L** of water per fill.

Desizing agent at 1 g/L = 3,000 × 1 = **3.0 kg**.

Cellulase enzyme at 0.8% on weight of garment (owg) = 300 × 0.008 = **2.4 kg**.

Pumice stone at 0.5 kg per kg of garment = 300 × 0.5 = **150 kg** (stones are screened, topped up for wear, and reused for several batches).

Doses are always stated in one of the two currencies — **g/L of bath** for bath-acting chemicals, **% owg** for garment-acting ones — and confusing them is the commonest batch-sheet error: 0.8% owg of enzyme is 2.4 kg here, but 0.8 g/L would be a different dose computed on 3,000 L.

5. Control points

A wash is approved against a **wash standard** (the buyer's approved washed garment) in standard light, checking shade, hand feel, and effect placement; the first batch of a style is a **bulk trial** whose result is signed before the rest follows. Shade is judged lot by lot because enzyme activity, stone condition and water quality drift; strength loss is the hidden cost of aggressive washing, and seam and pocket samples from each batch are the plant's evidence it stayed within limits. Effluent leaves through the plant's treatment plant — the cost appears in AM-4101 Topic 5.

Glossary

TERM	MEANING
Liquor ratio (LR)	Bath water volume per unit garment weight, e.g. 1:10 = 10 L per kg.
% owg	Dose as a percentage on the weight of the garments.
Desizing	Removal of weaving size/starch before dye-related washes.
Cellulase	Enzyme that hydrolyses surface cellulose, producing fading and softness.
Hydro-extractor	Centrifugal machine that removes bath water from washed garments before drying.

Key points and summary

- Washing cleans, softens, stabilises and fashions the garment; wet processes run in water, dry processes prepare effects on the dry garment.
- Enzyme + stone is the balanced standard wash; bleach and acid washes trade fibre strength for lightness and contrast.
- Sandblasting is excluded from modern practice on health grounds.
- Batch dosing is calculated in g/L or % owg — never mixed: the worked 300 kg batch needs 3,000 L per fill, 3.0 kg desizer, 2.4 kg enzyme, 150 kg pumice.

Practice

Answer in your own words; brief model answers follow.

1. A 250 kg batch is washed at LR 1:8 with enzyme at 1.0% owg and pumice at 0.4 kg/kg. Calculate water per fill, enzyme and pumice.
2. Why must desizing precede an enzyme wash on sized denim?

3. The recipe for the worked batch specifies enzyme at 0.8% owg. A worker instead weighs out 8 g/L on the 3,000 L bath. How much enzyme is used, and by what factor is the batch overdosed?
4. State the health reason sandblasting is banned, and name two dry processes that remain standard.

Answer key

1. Water = 250 × 8 = 2,000 L. Enzyme = 250 × 0.01 = 2.5 kg. Pumice = 250 × 0.4 = 100 kg.
2. Size coats the yarn surface; enzyme acting on a sized garment fades unevenly and weakly because it cannot reach the cellulose uniformly. Desizing exposes a clean, even surface.
3. Correct dose: 300 × 0.008 = 2.4 kg. Weighed out: 8 g/L × 3,000 L = 24 kg — a 10× overdose (24 ÷ 2.4), wasting chemical and over-fading the batch. The two currencies scale differently — % owg follows garment weight, g/L follows bath volume — so the recipe must state which it uses.
4. Silica dust from blasting causes silicosis, a disabling lung disease. Standard dry processes include whiskers/creasing, hand sanding, destroying, PP spray and resin creasing (any two).

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